How-To Library

Informatica

Deploy the Axon Data Governance Solution on the AWS Cloud Marketplace (7.1)

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Abstract

You can deploy Axon Data Governance on the Amazon Web Services (AWS) Marketplace. This deployment reference includes step-by-step instructions for deploying Axon Data Governance on the Amazon Web Services (AWS) Marketplace. It also includes information on prerequisites and how to troubleshoot common issues.

Supported Versions

- Axon Data Governance 7.1

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Overview

Automated reference deployments use AWS CloudFormation templates to launch, configure, and run the AWS compute, network, storage, and other services required to deploy a specific workload on AWS.

Intended Audience

This document is for users who are responsible for deploying Axon Data Governance on the AWS Marketplace.

As a user with administrator privileges to deploy applications on AWS, it is assumed that you are familiar with AWS resources such as Cloud formation, VPC, EC2, S3, RDS, Internet gateway, NAT Gateway, Route table, Security group, and Elastic IP.

As a user with administrator privileges to deploy applications on AWS, it is assumed that you are familiar with AWS resources such as Cloud formation, VPC, EC2, S3, Internet gateway, NAT Gateway, Route table, Security group, and Elastic IP.

Familiarity with the concepts of IP CIDR and public and private IP addresses is recommended.

You must also be familiar with Axon Data Governance.

For Axon Data Governance 7.1 documentation, see the Informatica Documentation Portal.

Specialized Knowledge

Before you deploy Axon Data Governance, we recommend that you become familiar with the following AWS services:

- Amazon VPC
- Amazon EC2
- Amazon S3
- Elastic IP Addresses

If you are new to AWS, see Getting Started with AWS.

Costs and Licenses

You are responsible for the cost of the AWS services used while running this deployment.

The AWS CloudFormation template for this deployment includes configuration parameters that you can customize. Some of these settings, such as instance type, will affect the cost of deployment. See the pricing pages for each AWS service you plan to use, for cost estimates.

This deployment requires a license for Axon Data Governance. To sign up for a license, contact Informatica.

The following table lists the instance types that are used for the deployment:

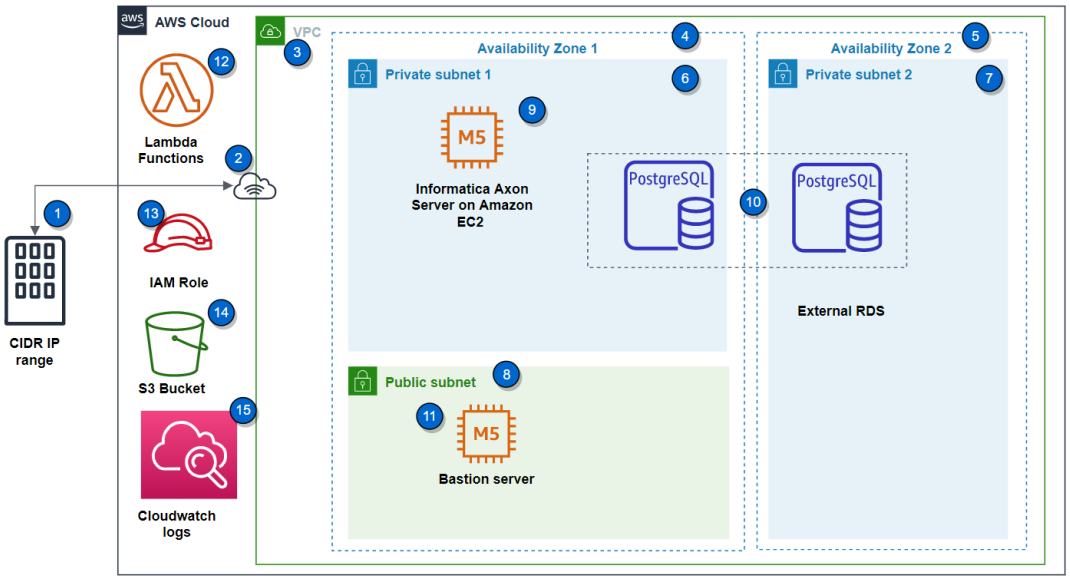
Virtual Machine	Instance Type
Informatica Axon Server	M5.2xlarge/M4.2xlarge
Bastion Server	M5.xlarge / M4.xlarge

Architecture

When you deploy the Axon Data Governance solution on a new Virtual Private Cloud (VPC) with the default parameters, the deployment builds the following Axon Data Governance environment on the AWS Marketplace:

- Informatica Axon on an EC2 instance.
- Informatica Axon Quick Look on a remote Windows bastion server, on a public subnet
- AWS security and account management services
- Lambda functions

The following image shows the architecture of Axon Data Governance on the AWS Marketplace:



The numbers in the architecture diagram correspond to items in the following list:

1. Your company or home network CIDR IP range.
2. A gateway to enter AWS VPC resources. It can be a NAT gateway, Internet gateway, or a virtual private gateway.
3. A virtual public cloud (VPC) configured in the Availability Zone to contain the Axon Data Governance deployment.
4. Availability Zone 1. This deployment provisions one private and one public subnet.
5. Availability Zone 2. This deployment provisions one private subnet.
6. Private subnet 1 to contain specific elements, such as Amazon EC2 and PostgreSQL databases, of the deployment. Creates one private subnet.
7. Private subnet 2 to contain specific elements, such as external PostgreSQL RDS, of the deployment. Creates one private subnet.
8. Public subnet to contain specific elements, such as bastion server, of the deployment. Creates one private subnet.
9. The Informatica Axon server.
10. External PostgreSQL RDS databases.
11. Informatica Axon Quick Look in a separate EC2 bastion server in a public subnet.
12. AWS Lambda functions.

13. IAM roles.
14. S3 storage that acts as a location for the license key.
15. Amazon CloudWatch logs.

AWS Resources in the Deployment

The deployment process creates or includes the following components:

- Components created when you specify the solution to create a VPC.
- Components created, as well as some that already exist, when you specify the solution to use an existing VPC.
- A bastion server, if you choose to create one.

Components in a Deployment on a New VPC

The deployment creates the following components when you deploy the software in a new virtual private cloud:

Component	Number of Components Created
VPC	1
Internet gateway	1
IAM role and instance profile	1
Subnets	3
Route tables	2
Security groups	The following security groups: - One security group for the Axon instance. - If you selected the option to create the bastion server, one security group to support remote login to the bastion server. - If you choose to create an external RDS, one security group for the external RDS instance.
Amazon EC2 instances	Informatica Axon with an EC2 instance and 100 GB of storage space. The instance created is of type m5.2xlarge or m4.2xlarge, depending on availability of the instance size in the selected region. Based on your choice, a bastion server with an EC2 instance that has 100 GB of storage space is created. The instance created is of type m5.xlarge or m4.xlarge, depending on availability of the instance size in the selected region.
Elastic IP addresses	One elastic IP address attached to the NAT gateway.
PostgreSQL RDS	Based on your choice, one PostgreSQL RDS instance with 100 GB storage space for external database configuration for Axon.

Components in a Deployment on an Existing VPC

The deployment creates the following components when you deploy the software in an existing virtual private cloud:

Component	Number of Components Created
Security groups	The following security groups: - One security group for the Axon instance. - If you selected the option to create the bastion server, one security group to support remote login to the bastion server. - If you choose to create an external RDS, one security group for the external RDS instance.
Amazon EC2 instances	Informatica Axon with an EC2 instance and 100 GB of storage space. The instance created is of type m5.2xlarge or m4.2xlarge, depending on availability of the instance size in the selected region. Based on your choice, a bastion server with an EC2 instance that has 100 GB of storage space is created. The instance created is of type m5.xlarge or m4.xlarge, depending on availability of the instance size in the selected region.
AWS Lambda functions	Three functions to confirm the validity of the VPC, the subnets, and the Informatica license.
IAM role and instance profile	1

Bastion Server

You can optionally deploy a bastion server as an element in the solution. The bastion server acts as a firewall between the internet and the cloud platform network where the solution deploys. It can also act as a remote Windows server which you can log into to run Informatica Axon Quick Look.

Informatica Axon Data Governance

The Axon application consists of the core Axon component, application services and microservices, and databases. These components exchange information through REST APIs.

Application Services

Axon uses application services and microservices to provide specific functionality that enable you to implement data governance. These services are independent, modular services that work together to enhance Axon capabilities.

Informatica® Axon Data Governance includes the following Axon application services:

Authorization

Provides access to Axon objects based on the segments that are assigned to users.

Bulk Upload

Uploads a large number of Axon objects and saves them to the PostgreSQL database.

Change Request

Provides a controlled and audited process to make changes to Axon objects. Axon uses the Camunda service to manage and run workflows within a change request.

Data Marketplace

Provides access to Axon Data Marketplace. In Axon Data Marketplace, you can search and access data collections that Informatica® Axon Data Governance users publish.

Document Upload

Attaches and maintains reference documents to an Axon object.

Notification

Sends notifications about changes to user roles, status of workflows and change requests, and status of bulk upload tasks.

Unison

Performs the Unison search. When you run a Unison search, Axon searches for the objects and the linked objects in all the available facets.

Value List

Creates or updates a large number of values for data set attributes and saves the values to the PostgreSQL database.

Informatica Axon Quick Look

Informatica Axon Quick Look is a client program that you can quickly access from the desktop. Use Axon Quick Look to search and look up glossary terms that are defined in Axon Data Governance. You can also select keywords from applications, such as Microsoft Word, Microsoft Excel, and Google Chrome, and use the hotkeys to perform a lookup in the Axon Quick Look application.

You can optionally deploy a bastion server as an element in the solution. The bastion server is a Windows instance installed with the Axon Quick Look client.

The bastion server has the following functions:

- Acts as a firewall to access the network.
- Acts as a remote Windows server that you can log in to run the Axon Quick Look client.

Before You Begin

Before you launch the automated deployment on the AWS Marketplace, verify the prerequisites and make the choices described in this section.

Network Prerequisites - AWS

Before you deploy Axon Data Governance, verify that you have completed the following network prerequisites for the Informatica application services.

Network prerequisites depend on whether you choose to deploy the solution to a new VPC or an existing VPC.

When you deploy to a new VPC, the automated deployment builds a new AWS environment consisting of the VPC, subnets, NAT gateways, security groups, and other infrastructure components. The deployment then deploys Axon Data Governance to the new VPC.

When you deploy to an existing VPC, the automated deployment provisions Axon Data Governance in your existing AWS infrastructure.

New VPC Prerequisites

Complete the following prerequisites when you choose to deploy into a new VPC:

- Ensure the selected Availability Zone has sufficient capacity to create a new subnet, route table, internet gateway, and NAT gateway.
- Ensure that the VPC creation has not exceeded the supported limit.

Existing VPC Prerequisites

Complete the following prerequisites when you choose to deploy into an existing VPC:

- Ensure that the VPC has DNS resolution enabled.
- You must also attach the subnet to a route table with a local route to the VPC CIDR.
- Verify that each subnet in the VPC is associated with a route table.
- Verify that the subnet associated with an internet gateway has the Auto-assign Public IPv4 property enabled.
- Verify that each route listed in the route table is defined as one of the following gateway types:
  - Internet gateway
  - NAT gateway
  - Virtual private gateway
- Verify that the subnet is associated with one of the following gateways:
  - Internet gateway
  - NAT gateway
  - Virtual private gateway
- Verify that the service subnets you choose are attached to a route table under the selected VPC.
- Verify that your VPC has a minimum of two subnets. If you choose to create a bastion server, you can deploy it to one of the two subnets or use a separate public subnet.

Subnets require an Internet gateway, an NAT gateway, or a virtual private gateway in their route tables, to allow the instances to download packages and software without exposing them to the Internet. Record the domain name configured in the DHCP options as explained in the Amazon VPC documentation. You provide the domain name in the VPC settings when you configure the deployment.

Account and Security Prerequisites

Complete the prerequisites in this section to verify that you obtained the required AWS account and private keys.

AWS Account and Region

If you do not already have an AWS account, create an account at aws.amazon.com.

Use the region selector in the navigation bar to choose the AWS region where you want to deploy Axon Data Governance on AWS.

Note: The region you choose determines the availability of certain sizes of EC2 instances.

If necessary, request a service limit increase for the EC2 instance type that you want to use for the Informatica domain. You might need to do this if you already have an existing deployment that uses this instance type, and you think you might exceed the default limit with this reference deployment.

Private Keys

Create a key pair in your preferred region.

When you log in to an EC2 instance, you use a password file for authentication. The file is called a private key file and has a file name extension of .pem.

If you do not have an existing .pem key to use, follow the instructions in the AWS documentation to create a key pair.

Note: Your administrator might ask you to use an existing key pair.

When you create a key pair, you save the .pem file to your desktop system. Simultaneously, AWS saves the key pair to your account. Make a note of the key pair that you want to use for the deployment, so that you can provide the key pair name during network configuration.

Account User Policies

Choose to grant the AWS user one of the following privileges:

- Administrator privileges
- The following minimum policies attached to the user account:
  - arn:aws:iam::aws:policy/AWSCloudFormationFullAccess
  - arn:aws:iam::aws:policy/AmazonEC2FullAccess
  - arn:aws:iam::aws:policy/CloudWatchFullAccess
  - arn:aws:iam::aws:policy/IAMFullAccess
  - arn:aws:iam::aws:policy/AmazonS3FullAccess
  - arn:aws:iam::aws:policy/AmazonRDSFullAccess
  - arn:aws:iam::aws:policy/AmazonVPCFullAccess
  - arn:aws:iam::aws:policy/AWSLambdaFullAccess

Storage Prerequisite

Create an Amazon S3 bucket for the Axon Data Governance license.

Amazon S3 stores objects that consist of one or more files and accompanying file metadata in buckets. For information about Amazon S3, see the AWS documentation.

To use the automated solution on the cloud marketplace, you must create an S3 bucket to store the license. You can store the license on the bucket level or in a folder within the S3 bucket.

Note the location of the license key and supply it when you configure the solution in the **License Key S3 Bucket URI** parameter.

Deploy Axon Data Governance on the AWS Marketplace

The automated deployment of Axon Data Governance on the AWS Marketplace uses the AWS CloudFormation console to guide your choices and launch the solution deployment.

You can select one of the following deployment options:

- Deploy Axon Data Governance into a new VPC (end-to-end deployment). This option builds and deploys a new AWS environment. The environment that it builds consists of the VPC, subnets, NAT gateways, security groups, and other infrastructure components, and then deploys Axon Data Governance into this new VPC.
- Deploy Axon Data Governance into an existing VPC. This option provisions Axon Data Governance in your existing AWS infrastructure.

The deployment also lets you configure additional settings such as CIDR blocks and Axon Data Governance settings, as discussed later in this guide.

Use the AWS CloudFormation console to specify the stack name and to enter parameter values to the template that you use to deploy the application. A stack is a logical representation of an AWS deployment.

Specify the identifiers for the VPC, for the subnet that contains the Amazon EC2 instances where the deployment process deploys the application components, and for the subnet where the deployment process creates the Model repository. The deployment process provisions the application components and creates an Axon server instance.

Availability Zones and Subnets

The automated solution deploys the components and services for Axon Data Governance into an Amazon VPC.

In the first subnet, the deployment process provisions Axon Data Governance on an Amazon EC2 instance within the VPC.

For more information about Axon Data Governance, see the Informatica Axon Administrator Guide and Informatica Axon User Guide.

Deployment Steps

You must perform multiple steps to deploy the Axon Data Governance 7.1 AWS Marketplace solution.

Step 1: Prepare an AWS Account

Perform the following steps to prepare an AWS account:

1. If you do not have an AWS account, create an account at http://aws.amazon.com by following the on-screen instructions.
2. Use the region selector in the navigation bar to choose the AWS Region where you want to deploy Axon Data Governance on AWS.
3. Create a key pair in your preferred region.
4. When you log in to an EC2 instance, you use a password file for authentication. The file is called a private key file and has a file name extension of .pem.
5. If you do not have an existing .pem key to use, follow the instructions in the AWS documentation to create a key pair.

   Note: Your administrator might ask you to use an existing key pair.

   When you create a key pair, you save the .pem file to your desktop system. Simultaneously, AWS saves the key pair to your account. Make a note of the key pair that you want to use for the Axon Data Governance instance, so that you can provide the key pair name during network configuration.
6. If necessary, request a service limit increase for the EC2 instance type that you want to use for the Informatica domain. You might need to do this if you already have an existing deployment that uses this instance type, and you think you might exceed the default limit with this reference deployment.

Step 2: Obtain a License for Axon Data Governance

Perform the following steps to obtain a license:

1. This deployment requires a license for Axon Data Governance. To sign up for a license, contact Informatica.
2. In your AWS account, create an S3 bucket.
3. Optionally, create a directory under the S3 bucket.
4. Place the Axon Data Governance license key file for the software in the S3 bucket or directory.

Step 3: Launch the Deployment

Each deployment takes approximately one to two hours to complete. You can choose to deploy Axon Data Governance on a new VPC or an existing VPC.

Perform the following steps to launch the deployment:

1. Log in to AWS Marketplace using the AWS account.
2. Type **Axon Data Governance** in the search box and press **Enter**.
3. Click **Axon Data Governance**. The **Product Overview** page appears.

4. Click **Continue to Subscribe**. The **Subscribe to this software** page appears.
5. Click **Continue to Configuration**. The **Configure this software** page appears.
6. Based on your requirement, select one of the following options from the **Delivery Method** drop-down box to specify where you want to deploy Axon Data Governance:
   - Axon Data Governance (New VPC). Select this option to deploy Axon Data Governance on a new VPC.
   - Axon Data Governance (Existing VPC). Select this option to deploy Axon Data Governance on an existing VPC.
7. Choose **7.1** from the **Software Version** list.
8. Click **Continue to Launch**. The **Launch this software** page appears.
9. Select **Launch CloudFormation** from the **Choose Action** drop-down list and click **Launch**. The **Create stack** page appears.
10. Check the region for the deployment displayed in the upper-right corner of the navigation bar, and change it if necessary. This is where the network infrastructure for Axon Data Governance will be built. The template is launched in the US West (Oregon) Region by default.
11. On the **Specify template** section, keep the default setting for the template URL, and then choose **Next**.

    The default option is `Amazon S3 URL`.
12. On the **Specify stack details** page, change the stack name if needed. Review the parameters for the template. Provide values for the parameters that require input.
13. Under **Capabilities**, select the check box to acknowledge that the template will create IAM resources.
14. To deploy the stack, click **Create**.

    Amazon AWS begins to create the stack and displays the CloudFormation dashboard. When the status is CREATE_COMPLETE, the deployment is complete. You can use the URL displayed on the **Output tab** to view the stack properties.
15. For all other parameters, review the default settings and customize them as necessary. When you finish reviewing and customizing the parameters, choose **Next.**

    In the following tables, parameters are listed by category and described separately for the two deployment options:
    - "Option1: Parameters for deployment into a new VPC" on page 11
    - "Option 2: Parameters for deployment into an existing VPC" on page 13

    Note: If you are deploying Axon Data Governance into an existing VPC, make sure that your VPC has a minimum of two subnets. If you choose a remote desktop, you can either use one of the two subnets or use a separate public subnet. These subnets require NAT gateways in their route tables, to allow the instances to download packages and software without exposing them to the Internet. You also need the domain name option configured in the DHCP options as explained in the Amazon VPC documentation. You'll be prompted for your VPC settings when you launch the deployment.

Option1: Parameters for deployment into a new VPC

Specify the identifiers for the VPC, for the subnet containing the Amazon EC2 instances where the deployment process deploys the application components, and for the subnets used by the Informatica services. The deployment process provisions the application components and creates the required repository databases based on the values you specify.

1. Enter the name of the stack to create. The stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).
2. Enter the parameters required to deploy the application.

The following table lists the parameters in the **Network Configuration** section:

Parameter	Default	Description
Availability Zone	Requires input	The Availability Zone to use for the subnet in the VPC.
VPC CIDR	10.0.0.0/16	Classless Inter-Domain Routing IP address range associated with the Amazon VPC. The CIDR range must have permissions to access the public subnet where the application runs. Format the string as `<IP address>/<IP network prefix>`. For example, to specify a range from `10.20.30.40` to `10.20.30.47`, enter `10.20.30.40/29`.
Public Subnet CIDR	Requires input	CIDR block for the public subnet. Ensure that you include the subnet in the VPC CIDR.
Private Subnet 1 CIDR	10.0.0.0/19	CIDR block for the private subnet 1. Ensure that you include the subnet in the VPC CIDR.
Private Subnet 2 CIDR	10.0.192.0/21	CIDR block for the private subnet 2. Additional subnet is required for the external RDS deployment.
CIDR IP Range	Requires input	The CIDR IP range that is permitted to access Axon Data Governance. To reduce the potential of inbound attacks from unknown IP addresses, you can use a constrained CIDR range to reduce. For example, to specify the range of 10.20.30.40 to 10.20.30.47, enter 10.20.30.40/29.

The following table lists the parameters in the **Bastion Server for Informatica** section:

Parameter	Default	Description
Deploy Bastion Server	Requires input	Select **Yes** to deploy a bastion server to access other resources in the VPC.

The following table lists the configuration parameters for the **External PostgreSQL RDS** database:

Parameter	Default	Description
Deploy PostgreSQL RDS	No	Select Yes to deploy an external PostgreSQL RDS database.
DB admin username	adminuser	The Admin user name for the database instance.

Parameter	Default	Description
DB admin password	Requires input	The Admin password for the database instance.
DB Multiple Availability Zones	false	Specifies if the database instance is deployed to multiple Availability Zones for High Availability.

The following table lists the parameters in the **Amazon EC2 Configuration** section:

Parameter label (name)	Default	Description
Key Pair Name	Requires input	Name of an existing Amazon EC2 key pair to enable SSH and Remote Desktop Protocol (RDP) access to Axon Data Governance.

The following table lists the parameters in the **Informatica Axon Data Governance** section:

Parameter label (name)	Default	Description
License Key S3 Bucket URI	Requires input	Required. The S3 bucket URI for the Axon Data Governance license key in the following format: `s3://<bucket-name>/<directory>/<license key>`
Download Resources from S3 Bucket	No	Choose whether to use local VM stored configuration files or download from an S3 bucket owned by Informatica.

3. After you configure the parameters in the **Informatica Axon Data Governance** section, click **Next**.
4. On the **Options** page, expand the **Advanced** tab.
5. Set the **Rollback on Failure** property to **No**.
6. Click **Next**.
7. Click **Create**.
8. When the status for the stack reads CREATE_COMPLETE, click the stack name.

Option 2: Parameters for deployment into an existing VPC

1. Enter a name for the stack. The stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).
2. Enter the parameters required to deploy the application.

The following table lists the parameters in the **Network Configuration** section:

Parameter	Default	Description
VPC	Requires input	ID of your existing VPC where you want to deploy Axon Data Governance.
Subnet 1	Requires input	Select the subnet ID where you want to deploy Axon Data Governance. Make sure that the subnet is included in the VPC and associated with an Internet Gateway or a NAT Gateway. Ensure that service subnet ID1 and service subnet are not the same.
Subnet 2	Requires input	CIDR block for the private subnet 2. Additional subnet is required for the external RDS deployment.
CIDR IP Range	Requires input	The CIDR IP range that is permitted to access Axon Data Governance. To reduce the potential of inbound attacks from unknown IP addresses, you can use a constrained CIDR range. For example, to specify the range of 10.20.30.40 to 10.20.30.47, enter 10.20.30.40/29.
Pre-validation Check	Requires input	Select to validate the VPC and subnets before deploying Axon Data Governance. If you choose to have the automated solution perform a validation check before launching the deployment, the check validates your parameter inputs. If the deployment as specified fails the check, the deployment process stops. The deployment logs record the problem.

The following table lists the parameters in the **Bastion Server for Informatica** section:

Parameter	Default	Description
Deploy Bastion Server	Requires input	Select **Yes** to deploy a bastion server to access other resources in the VPC.
Bastion Subnet	Requires input	Select the subnet for the bastion server if you selected **Yes** for the previous option. Ensure that the subnet is included in the VPC that you use and attached to an Internet Gateway.

The following table lists the configuration parameters for the **External PostgreSQL RDS** database:

Parameter	Default	Description
Deploy PostgreSQL RDS	No	Select Yes to deploy an external PostgreSQL RDS database.
DB admin username	adminuser	The Admin user name for the database instance.
DB admin password	Requires input	The Admin password for the database instance.
DB Multiple Availability Zones	false	Specifies if the database instance is deployed to multiple Availability Zones for High Availability.

The following table lists the parameters in the **Amazon EC2 Configuration** section:

Parameter label (name)	Default	Description
Key Pair Name	Requires input	Name of an existing Amazon EC2 key pair to enable SSH and Remote Desktop Protocol (RDP) access to Axon Data Governance.

The following table lists the parameters in the **Informatica Axon Data Governance** Configuration section:

Parameter label (name)	Default	Description
License Key S3 Bucket URI	Requires input	The S3 bucket URI for the Axon Data Governance license key in the following format: `s3://<bucket-name>/<directory>/<license key>`
Download Resources from S3 Bucket	No	Choose whether to use local VM stored configuration files or download from an S3 bucket owned by Informatica.

3. Click **Next** after you configure the parameters in the Informatica Axon Data Governance Configuration section.
4. On the **Configure stack options** page, you can specify tags (key-value pairs) for resources in your stack and set advanced options. When you are done, choose **Next**.
5. On the **Review** page, review and confirm the template settings. Under Capabilities, select the check box to acknowledge that the template will create IAM resources.
6. Choose **Create** to deploy the stack.
7. Monitor the status of the stack. When the status is `CREATE_COMPLETE`, the deployment is complete.

See the Axon Data Governance User Guide for information about logging in to Axon Data Governance.

Monitoring Instance Provision and Informatica Axon Creation

You can use cloud platform dashboards, logs, or other artifacts to see whether cluster creation succeeded and how to locate and identify Axon Data Governance on the cloud platform.

Deployed AWS Resources

You can monitor the progress of the solution deployment in the Create Stack section of the AWS portal. Each row of the display shows stack components and creation status. The stack is complete when the status reads CREATE_COMPLETE.

When you deploy the solution to an existing VPC, the console displays the following stack components:

Component	Description
StackName/AxonStack	The main stack that is created by default.
StackName-LicenseCheckLambdaStack	Checks whether the license key file is valid. Created by default.
StackName-BastionSubnetCheckLambdaStack	Checks whether the bastion server subnet is valid. Created by default. Lambda function will be created only when: - Pre-validation check parameter set to Yes - Deploy Bastion Server set to Yes
StackName-PublicIpDnsCheckLambdaStack	Checks whether an auto-assign public IPv4 address is enabled for a subnet. Created by default.
StackName-SubnetCheckLambdaStack	Checks whether the first subnet is valid. Created by default. Creates the Lambda function if the Pre-validation Check parameter is set to Yes.
StackName-IAMStack	Creates IAM roles and profiles used by EC2 instances. Created by default.
StackName-SecurityGroupStack	Creates security groups and network interfaces used by EC2 instances. Created by default.
StackName-RDSStack	Create a PostgreSQL database instance for an external PostgreSQL module for Axon. Created if the **Deploy PostgreSQL RDS** parameter is set to **Yes**.
StackName-AxonServerStack	Creates an Informatica gateway node on which Axon Data Governance runs on an EC2 instance. Created by default.
StackName-BastionServerStack	Creates bastion server on an EC2 instance. Created when you choose to create the bastion server.

When you deploy to a new VPC, the console displays the following stack components:

Component	Condition for Creation
StackName-VPCStack	Creates network-related resources including the VPC, subnets, internet gateway, and the NAT gateway. Created by default.
AxonStack	The main stack. Created by default.

Output Tab Properties

Click the **Output** tab to view the output parameters after the stack is successfully launched.

The following table lists the outputs:

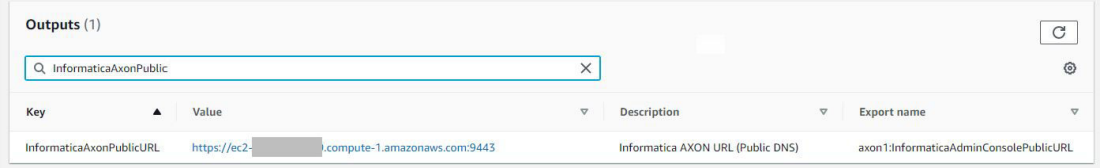
Output	Description
AxonUser	User name of the Axon user.
AxonPassword	Password for the Axon user.
BastionPrivateDNS	DNS of the private bastion server. For example: ip-<IPv4 address>-<region designation>.compute.internal
BastionPublicIPDNS	DNS of the public bastion server. For example: ip-<IPv4 address>-<region designation>.compute.amazonaws.com
AxonQuickLookInstaller	The download location for the Axon Quick Look client installer.
InformaticaAxonURL	Private link with the port for the Axon application hosted on a private server.
InformaticaAxonPublicURL	Public link with the port for the Axon application.
InformaticaAxonLogs	Path to the Axon installation log.
InstanceSetupLogs	Path to the domain EC2 host setup log.
RDSEndpoint	The PostgreSQL RDS instance host name created by deployment. Available if the **Deploy PostgreSQL RDS** parameter is set to **Yes**.

For the Axon Data Governance administrator tasks, see the Informatica Axon Administrator Guide.

Deployed Informatica Resources

The list of deployed resources includes the **InformaticaAxonPublicURL**. Click the linked value of this property.

The following image shows an example of this property, with the IP address portion redacted:



You can access the Axon application from the deployment output section.

If you are within same AWS virtual network, you can directly access the Axon application.

If you are outside of the AWS virtual network, perform the following tasks:

1. Open the `hosts` file from the following location:
   `C:\Windows\System32\drivers\etc\hosts file`
2. Enter the host and DNS information in the following format:
   `<public IP>space<Public IPv4 DNS>space<Private IPv4 DNS>`

Solution Deployment Log Files

You can review the following log files to see the solution deployment as it happened.

Review the following log files in the `/var/log` directory:

- cfn-init.log
- cfn-init-cmd.log
- cfn-wire.log
- cloud-init.log
- messages

Review the following log files:

- `/opt/Infa_node_configure.log`
- `/opt/Infa_OneClick_Solution.log`
- `/opt/Informatica/AxonServer/Configure.log`
- `/opt/Informatica/AxonServer/Informatica_Axon_7.1.0.log`

Informatica Axon Data Governance Logs

You can access the following logs on the Axon VM:

Command execution log

This log records the creation of Informatica Axon Data Governance.

At the top of the log file is a summary section that lists automated tasks and their status. Below the summary section are detailed sections about each task. If any of the tasks fail to complete, you can look at the detailed section for the task to troubleshoot.

Review the following log files:

- `/opt/Infa_node_configure.log`
- `/opt/Infa_OneClick_Solution.log`
- `/opt/Informatica/AxonServer/Configure.log`
- `/opt/Informatica/AxonServer/Informatica_Axon_7.1.0.log`

CloudWatch logs for Informatica

CloudWatch is an AWS service that enables you to monitor all of the events in your AWS deployments. CloudWatch provides a central location from which to monitor EC2 instances and other resources. for more information, see the AWS documentation.

To monitor Axon Data Governance events in CloudWatch, go to the CloudWatch page and search for "Informatica."

If no log files are returned by the search, log into the VM where the Informatica domain is installed and check for logs with the filename "`cfn*.log`"

Access the Informatica deployment logs through the AWS console:

1. In the Amazon Web Services Console, select CloudWatch from the Services menu.
2. In the navigation pane, choose **Logs** > **Log groups**.
3. Enter InformaticaAxonLogGroup in the search pane.

CloudWatch logs for Lambda functions

If the automated deployment fails during execution, CloudWatch logs display information about process errors.

Access the CloudWatch logs through the AWS console:

1. In the Amazon Web Services Console, select CloudWatch from the Services menu.
2. In the navigation pane, choose **Logs** > **Log groups**.
3. Enter `/aws/lambda/<stack name>` in the search pane.
4. Select the log group associated with the name of the stack that failed.

Troubleshooting

Review the AWS log files to troubleshoot issues with your Axon Data Governance deployment.

Stack Failure

Review the following log files if an EC2 instance where Informatica services run is running but the stack fails, or if the Axon application fails to launch:

- Node Configuration log file: `/opt/Infa_node_configure.log`
- Installation log file: `/home/Infa_OnceClick_Solution.log`
- Command Initialization log file: `/var/log/cfn-init-cmd.log`
- Service Initialization log file: `/var/log/cfn-init.log`
- Cloud log file: `/var/log/cloud-init.log`

Stack Launch Failure When the Pre-validation Check Parameter is Enabled

Review the Lambda CloudWatch log files if a stack failure occurs when the Pre-validation Check option is enabled.

1. In the AWS Management Console, select **CloudWatch** from the **Services** menu.
2. Select **Logs**, and select **Log Groups**.
3. Enter the following string in the search bar:

   `/aws/lambda/<stack name>`
4. Click the log group associated with the stack.

Stack Launch Failure When Rollback Is Enabled

Review the Lambda CloudWatch log files if a stack failure occurs when the **Rollback on Failure** option is enabled.

1. In the AWS Management Console, select **CloudWatch** from the **Services** menu.
2. Select **Logs**, and select **Log Groups**.

3. Enter the following string in the search bar: `InformaticaAxonLogGroup`
4. Click a log file in the log group associated with the stack.

   You might want to review the following log files:

   - installation.log
   - cfn-init
   - AXON-Log
   - AXON-Configure
   - AXON-ErrorLog

Frequently Asked Questions

Q. I encountered a CREATE_FAILED error when I launched the deployment. What should I do?

A. If AWS CloudFormation fails to create the stack, we recommend that you relaunch the template with **Rollback on failure** set to **No**. (This setting is under **Advanced** in the AWS CloudFormation console, **Options** page.) With this setting, the stack's state will be retained, and the instance will be left running, so you can troubleshoot the issue. (You will want to look at the log files in `/var/log/cfn-init-cmd.log`)

Note: When you set **Rollback on failure** to **No**, you will continue to incur AWS charges for this stack. Please make sure to delete the stack when you have finished troubleshooting.

For additional information, see Troubleshooting AWS CloudFormation on the AWS website or contact us on the AWS Discussion Forum.

Q. I encountered errors while installing Axon Data Governance. Where can I find more information?

A. View the `/opt/Infa_OneClick_Solution.log` log file to see more information about the errors that you encountered.

Q. I encountered a size limitation error when I deployed the AWS CloudFormation templates.

A. We recommend that you launch the deployment templates from the location we provided or from another S3 bucket. If you deploy the templates from a local copy on your computer or from a non-S3 location, you might encounter template size limitations when you create the stack. For more information about AWS CloudFormation limits, see the AWS documentation.

Q. What should I do before I relaunch the template with **Rollback on failure**?

A. Ensure that you disable rollback when you launch the stack in case of a stack failure. If Axon Data Governance server stack is created, check the following logs to identify the issues:

- `/opt/Infa_node_configure.log`
- `/opt/Infa_OneClick_Solution.log`
- `/var/log/cfn-init.log`

If the Axon Data Governance server stack is not created, you can check the CloudWatch logs to identify the issues. For more information, see the "Stack Launch Failure When Rollback Is Enabled" on page 19 section.

Additional Resources

AWS Services

Refer to the following AWS Services documentation:

- AWS CloudFormation
- Amazon EBS
- Amazon EC2
- Amazon VPC

Axon Data Governance Documentation

Refer to Axon Data Governance at Product Documentation.

Reference Deployment

Refer to the AWS deployment at AWS Deployment home page.

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